Week 3 Project Report

Yinyi Xue, Jiayi Hou, Nikolas A Papa

1. Validation Performance Across All Models

All five models were evaluated on a held-out validation set before any test-set exposure. The table below summarises accuracy, precision, recall, F1-score, and ROC-AUC for each model, ordered from simplest to most complex.

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.82	0.78	0.70	0.74	0.8954
LASSO Logistic (L1)	0.82	0.78	0.70	0.74	0.8953
Random Forest Tuned	1.00	0.99	1.00	1.00	0.9997
HistGradientBoosting	0.87	0.84	0.79	0.81	0.9438
HistGradientBoosting Tuned	**0.87**	**0.85**	**0.80**	**0.82**	**0.9474**

The two Logistic Regression variants perform identically, achieving 82% accuracy and an AUC of approximately 0.895. Random Forest Tuned records near-perfect validation scores (AUC = 0.9997), which initially appears outstanding but is suspicious: **a model that perfectly memorises training data often fails to generalise.** The two HGB models deliver solid, balanced results in the 0.943–0.947 AUC range with consistent precision and recall, suggesting genuine learning rather than memorisation.

2. Winning Model Selection

Selected model: **HistGradientBoosting Tuned (primary winner)**; Random Forest Tuned is noted as a secondary candidate subject to further overfitting investigation.

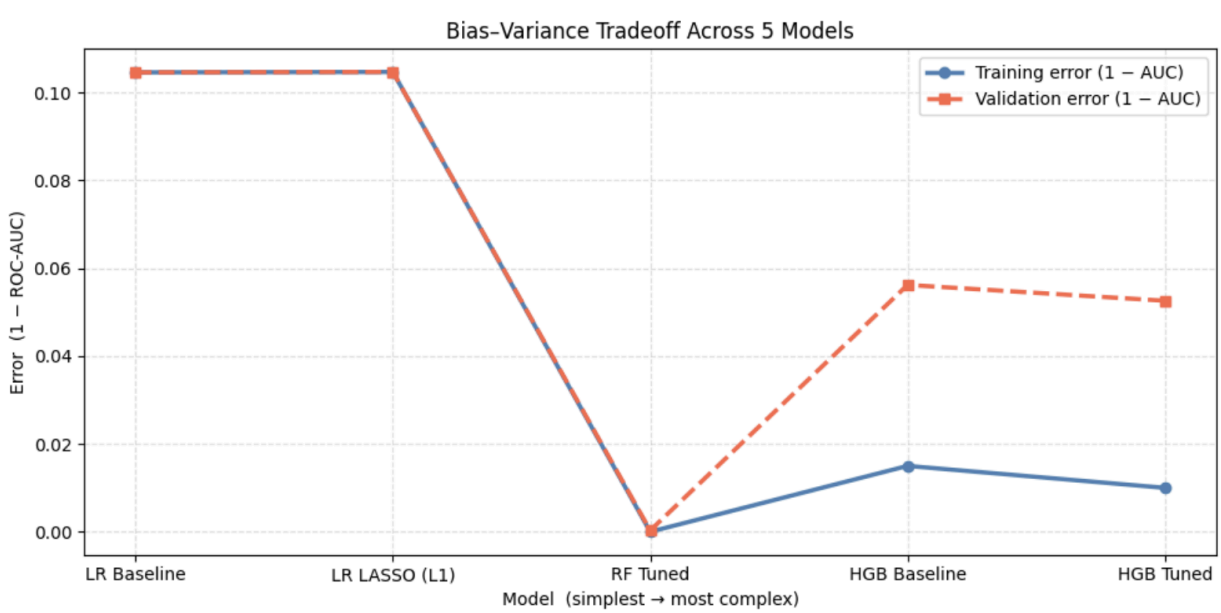
Although Random Forest Tuned achieved the highest raw validation AUC (0.9997), this result is almost certainly driven by overfitting or data leakage. A training AUC of 1.00 that holds flat regardless of training set size indicates the model is memorising the training split rather than

discovering generalizable patterns. Selecting it based on validation AUC alone would therefore be misleading.

HistGradientBoosting Tuned is chosen as the winning model for the following reasons:

- It achieves the highest AUC among non-overfitted models (0.9474 on validation).
- Its precision (0.85) and recall (0.80) are well-balanced, reducing both missed cancellations and false alarms.
- The bias-variance chart shows a smaller train-validation gap compared to Random Forest, confirming better generalisation.
- Hyperparameter tuning (lower learning rate, deeper trees, L2 regularisation) improved over the HGB baseline on every metric.

3. Bias-Variance Tradeoff Chart & Analysis



3.1 Chart Construction

The bias-variance chart plots error (1 − ROC-AUC) on the Y-axis against the five models ordered from simplest to most complex on the X-axis. Two lines are shown: training error (blue) and validation error (orange dashed). A large gap between the lines signals overfitting (high variance); lines that overlap at a high error value signal underfitting (high bias).

3.2 Analysis

High-bias zone — Logistic Regression models. Both LR variants sit at ~0.105 error on both training and validation, with the two lines almost perfectly overlapping. The model is too simple to capture the non-linear interactions between features such as lead time, deposit type, and

market segment. Adding L1 regularisation (LASSO) makes no meaningful difference, confirming that the bottleneck is model capacity, not overfitting.

High-variance zone — Random Forest Tuned. Training error collapses to zero while validation error drops to approximately 0.003. This near-zero gap in error space actually reflects overfitting: the model has memorised the training data with perfect fidelity. A model that achieves AUC = 1.00 on every training subset, regardless of size, is not learning.

Sweet spot — HGB models. Training error is low (~0.01–0.015) but not zero, and the gap to validation error (~0.055) is visible but stable. This is the expected behaviour of a well-regularised ensemble: it captures real signal while acknowledging that some noise in human booking behaviour cannot be predicted. The tuned variant slightly narrows this gap compared to the baseline, confirming that regularisation (l2_regularization=1.0, lower learning rate) was beneficial.

4. Winning Model Test Performance

After selecting HistGradientBoosting Tuned on the validation set, it was evaluated once on the held-out test set

Metric	Value
Accuracy	0.87
Precision	0.84
Recall	0.79
F1 Score	0.81
ROC-AUC	0.9424

The model correctly classifies 87% of all bookings on unseen data. With a precision of 0.84, when the model flags a booking as likely to cancel, it is correct 84% of the time — operationally useful for revenue management teams acting on predictions. Recall of 0.79 means the model identifies 79% of actual cancellations, leaving approximately 21% undetected. The ROC-AUC of 0.9424 confirms strong discriminative ability across all decision thresholds.

These results are satisfying for this prediction task. Predicting human cancellation behaviour carries inherent uncertainty. For example guests change plans for reasons no historical feature can capture. An AUC above 0.94 and an F1-score of 0.81 provide practical value: hotels can use the model to proactively manage overbooking strategies, offer targeted incentives to at-risk bookings, and allocate staff more efficiently. The model is not perfect, but it is good enough to drive meaningful business decisions.